

On $\bar{\chi}(G) - \alpha(G) > 0$ gap recognition and $\alpha(G)$ -upper bounds

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Abstract

We show that for a graph G it is NP -hard to decide whether its independence number $\alpha(G)$ equals its clique partition number $\bar{\chi}(G)$ even when some minimum clique partition of G is given. This implies that any $\alpha(G)$ -upper bound provably better than $\bar{\chi}(G)$ is NP -hard to compute.

To establish this result we use a reduction of the Quasigroup Completion Problem (QCP, known to be NP -complete) to the maximum independent set problem. A QCP instance is satisfiable if and only if the independence number $\alpha(G)$ of the graph obtained within the reduction is equal to the number of holes h in the QCP instance. At the same time, the inequality $\bar{\chi}(G) \leq h$ always holds. Thus, QCP is satisfiable if and only if $\alpha(G) = \bar{\chi}(G) = h$. Computing the Lovász number $\vartheta(G)$ we can detect QCP unsatisfiability at least when $\bar{\chi}(G) < h$. In the other cases QCP reduces to $\bar{\chi}(G) - \alpha(G) > 0$ gap recognition, with one minimum clique partition of G known.

Keywords: independence number, clique partition number, Lovász number, latin square, quasigroup completion problem.

1 Introduction

Let $G(V, E)$ be a simple undirected graph. An *independent set* of vertices is a subset $S \subseteq V$ such that any two vertices of S are *not* adjacent. The *maximum independent set problem* asks for an independent set of the maximum cardinality. This cardinality $\alpha(G)$ is called the *independence number* of the graph, and is NP -hard to compute [1]. A *clique* Q is a subset of V such that any two vertices of Q are adjacent. The *minimum clique partition problem* asks for a smallest by cardinality set of cliques $\{Q_1, \dots, Q_{\bar{\chi}}\}$ containing every vertex $v \in V$ in exactly one of the cliques. The cardinality $\bar{\chi}(G)$ of this set is called the *clique partition number*. It is equal to the *chromatic number* $\chi(\bar{G})$ (minimum number of vertex colors needed to provide different colors for any pair of adjacent vertices) of the complementary graph. The minimum clique partition problem is also NP -hard [1]. Obviously, the inequality $\alpha(G) \leq \bar{\chi}(G)$ holds as no two vertices of an independent set can belong to the same clique.

There exists a polynomial-time computable function $\vartheta(G)$ “sandwiched” between those two NP -hard numbers [3, 6]:

$$\alpha(G) \leq \vartheta(G) \leq \bar{\chi}(G). \quad (1)$$

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One simple definition of $\vartheta(G)$ is via minimum of the largest eigenvalue of so-called *feasible matrices* $A = (a_{ij})_{n \times n}$:

$$\vartheta(G) = \min_A \lambda_{\max}(A), \quad (2)$$

$$\text{s.t. } a_{ij} = 1 \text{ if } (i, j) \notin E; \quad a_{ij} = a_{ji}$$

(that is, to obtain $\vartheta(G)$ we minimize the largest eigenvalue of a symmetric matrix having 1's on the main diagonal and in all entries corresponding to non-edges, while the other entries are arbitrary). $\vartheta(G)$ is called the *Lovász number (or ϑ -function)* of a graph. It serves as an upper bound for the independence number and as a lower bound for the clique partition number simultaneously. Besides, there are increasingly tight sequences of polynomial-time computable upper bounds for $\alpha(G)$ based on “lift-and-project” method [5] and the concept of matrix copositivity [11].

A *latin square* is an $n \times n$ matrix filled with integers from 1 to n so that each number occurs exactly once in any row and in any column. One example is $L = (\ell_{ij})_{n \times n}$ such that

$$\ell_{ij} = ((i + j - 2) \bmod n) + 1. \quad (3)$$

In the *Quasigroup Completion Problem* (QCP, a.k.a. latin square completion) we are given an $n \times n$ array partially filled with integers from $\{1 \dots n\}$ and it is asked whether there is a completion for all h empty cells (*holes*) such that it gives a latin square. QCP is *NP*-complete [4]. Recently it has been intensively studied, especially from constraint programming and boolean satisfiability viewpoints [7, 8, 9, 10, 12, 13, 14].

In this paper we show a reduction of QCP to the maximum independent set problem. The obtained graph instances obey $\alpha(G) \leq h$ and $\bar{\chi}(G) \leq h$ constraints. At that, the original QCP instance is satisfiable if and only if $\alpha(G) = \bar{\chi}(G) = h$. This allows us to obtain some results restricting polynomial-time recognition of $\bar{\chi}(G) - \alpha(G) > 0$ gap and computation of tight upper bounds on $\alpha(G)$ under $P \neq NP$ assumption. In fact, unless $P = NP$, it means there is no polynomial-time computable upper bound on $\alpha(G)$ provably better than $\vartheta(G)$ and, in turn, $\vartheta(G)$ does not bound $\alpha(G)$ provably better than $\bar{\chi}(G)$.

2 Latin square 3D encoding

The concept of latin square can be also expressed via a three-dimensional array of 0-1 values. Namely, let a 0-1 variable x_{ijk} , $i, j, k \in \{1 \dots n\}$ denote “Cell (i, j) is filled with number k ”. The array of these variables determines a latin square if and only if

$$\begin{cases} \forall i, j \quad \sum_{k=1}^n x_{ijk} = 1, \\ \forall i, k \quad \sum_{j=1}^n x_{ijk} = 1, \\ \forall j, k \quad \sum_{i=1}^n x_{ijk} = 1. \end{cases} \quad (4)$$

These conditions correspond to maximum independent sets of a graph, whose vertices are triples (i, j, k) and there is an edge between two of them if and only if two of their entries coincide. This graph Γ is known as $H(3, n)$ Hamming graph, see e.g. [2].

Lemma 1 $\alpha(\Gamma) = n^2$. *There is a one-to-one correspondence between maximum independent sets of Γ and $n \times n$ latin squares.*

Proof. First, we prove Γ does not have an independent set larger in size than n^2 . Indeed, there are only n^2 distinct pairs of two first entries (i, j) for the vertices $\{(i, j, k)\}$. Thus, in any vertex subset X such that $|X| > n^2$ there is at least one vertex pair $((i, j, k), (p, q, r))$ such that $(i, j) = (p, q)$. This vertex pair must be connected by an edge. So, X is not an independent set.

Now, consider a latin square $L = (\ell_{ij})_{n \times n}$ and the vertex subset $S = \{(i, j, k) : \ell_{ij} = k\}$. It contains n^2 vertices because there are n^2 distinct (i, j) pairs. Let $(i, j, k) \in S$ and $(p, q, r) \in S$ be two distinct vertices. As $i = p$ and $j = q$ would have implied $k = r$ by the definition of S , this case is not possible. Thus, if $i = p$, then $j \neq q$ and $k \neq r$ as L does not have two equal numbers on the same row. Similarly, if $j = q$, then $i \neq p$ and $k \neq r$ as L does not have two equal numbers on the same column. Therefore, there are no triples in S with exactly two common entries and, hence L defines a maximum independent set of Γ . As (3) provides a latin square for any $n > 0$, one obtains $\alpha(\Gamma) = n^2$ as claimed.

Conversely, it is easy to see that any Γ maximum independent set $S = \{(i, j, k)\}$, $|S| = n^2$ defines a latin square $L = (\ell_{ij})_{n \times n}$ such that $\ell_{ij} = k$ if and only if $(i, j, k) \in S$. QED.

To reduce QCP to the maximum independent set problem we will use subgraphs of Γ . Let the QCP input be a matrix $L = (\ell_{ij})_{n \times n}$ such that $\ell_{ij} = k \in \{1 \dots n\}$ if the cell (i, j) is prefilled with k , and $\ell_{ij} = 0$ otherwise. Correspondingly, the number of holes h is the total number of entries (i, j) such that $\ell_{ij} = 0$. Without loss of generality we assume that this input does not immediately violate the latin square constraints. That is, $\ell_{ij} = \ell_{iq} > 0$, $j \neq q$ or $\ell_{ij} = \ell_{pj} > 0$, $i \neq p$ cases never occur. Otherwise, the QCP instance is trivially unsatisfiable. Define a graph $G(V, E)$ with vertices

$$V = \{(i, j, k) : (\ell_{ij} = 0) \& (\forall p : \ell_{pj} \neq k) \& (\forall q : \ell_{iq} \neq k)\}. \quad (5)$$

As earlier, put an edge between distinct vertices (i, j, k) and (p, q, r) when they have two common entries:

$$E = \{((i, j, k), (p, q, r)) : (i = p) \& (j = q) \& (k \neq r) \vee (i = p) \& (j \neq q) \& (k = r) \vee (i \neq p) \& (j = q) \& (k = r)\}. \quad (6)$$

In other words, $G(V, E)$ is the subgraph of Γ induced by non-neighbors of those vertices (i, j, k) for which $\ell_{ij} = k > 0$.

Lemma 2 $\alpha(G) \leq h$. *The QCP instance given by the matrix L is satisfiable if and only if $\alpha(G) = h$.*

Proof. Let $S_0 = \{(i, j, k) : \ell_{ij} = k > 0\}$ be the vertex subset of Γ corresponding to the partial completion given by L . Obviously, $|S_0| = n^2 - h$. Since the partial completion obeys the latin square constraints, S_0 is an independent set. Denote by $N^+(S_0)$ the closed neighborhood of S_0 , that is, union of S_0 with the set of Γ vertices adjacent to at least one vertex from S_0 . $G(V, E)$ is obtained by removing $N^+(S_0)$ from Γ .

Assume $G(V, E)$ has an independent set S_1 of size greater than h . Then $S_0 \cup S_1$ is an independent set of Γ having more than n^2 vertices, contradicting Lemma 1. Hence $\alpha(G) \leq h$.

Let $G(V, E)$ have a maximum independent set S_1 of size h . Then $S_0 \cup S_1$ is a maximum independent set of Γ , so S_1 determines a correct completion of the QCP input to a latin square. Therefore, the given QCP instance is satisfiable. Conversely, if the given input matrix L admits a completion to a latin square, we can take the maximum independent set S of Γ corresponding

to this latin square and observe that $S \setminus S_0$ is an independent set of $G(V, E)$ of size h . Therefore, the QCP instance is satisfiable if and only if $\alpha(G) = h$. QED.

Thus, we have described a reduction of QCP to the maximum independent set problem on Γ subgraphs. In the next section we present the results based on clique partition of these subgraphs.

3 The main results

Lemma 3 *Let $G(V, E)$ be a graph obtained within the QCP reduction to the maximum independent set problem. Then $\bar{\chi}(G) \leq h$.*

Proof. Let $L = (\ell_{ij})_{n \times n}$ be the QCP input matrix as described above. V may include only such vertices (i, j, k) for which $\ell_{ij} = 0$. We note that all $(i, j, k) \in V$ corresponding to one hole $\ell_{ij} = 0$ comprise a clique. Hence V is a union of not more than h of such cliques. This implies $\bar{\chi}(G) \leq h$. QED.

Therefore, computing the Lovász number $\vartheta(G)$ on the described graphs we can efficiently detect QCP unsatisfiability at least when $\bar{\chi}(G) < h$. We may say that the inequality $\vartheta(G) < h - \epsilon$ for some fixed $0 < \epsilon < 1$ designates an easily recognizable subclass of unsatisfiable QCP instances. In the other cases, QCP is equivalent to deciding whether $\alpha(G) = \bar{\chi}(G)$ provided $\bar{\chi}(G) = h$ and the clique partition defined in the proof of Lemma 3 is a minimum one. Thus, we have deduced the following:

Theorem 1 *For a graph G is it NP-hard to decide whether there is a gap between its independence and clique partition numbers $\bar{\chi}(G) - \alpha(G) > 0$ provided some minimum clique partition of G is given.*

We note that currently we are not aware of any graph G obtained within the reduction from an unsatisfiable QCP instance for which $\vartheta(G) = \bar{\chi}(G) = h$.

Corollary 1 *For a graph G is it NP-hard to decide whether there is a gap between its independence and clique partition numbers $\bar{\chi}(G) - \alpha(G) > 0$.*

Though it immediately follows from Theorem 1, there is also a simple direct proof of this fact. Assume we have an oracle answering whether $\bar{\chi}(G) - \alpha(G) > 0$ for any graph G . Define G_i as the graph composed of G and i additional mutually independent vertices, each of which is connected with every vertex of G . Note that $\alpha(G_i) = \max(\alpha(G), i)$ and $\bar{\chi}(G_i) = \max(\bar{\chi}(G), i)$. Submit the graphs G_i , $i = 0, 1, \dots$ to the oracle until it says $\alpha(G_i) = \bar{\chi}(G_i)$. There cannot be more than $\bar{\chi}(G)$ of such queries. Upon termination, $\vartheta(G_i) = \bar{\chi}(G)$ since $\alpha(G_i) = \bar{\chi}(G_i) = \bar{\chi}(G)$, so using the oracle we can compute $\bar{\chi}(G)$ in polynomial time. (In fact, we have to compute $\vartheta(G_i)$ only if the process stops with $i = 0$, that is, when $\alpha(G) = \bar{\chi}(G)$. Otherwise the terminal value i gives $\bar{\chi}(G)$.)

Corollary 2 *Unless $P = NP$, there is no polynomial-time computable upper bound on the independence number $\alpha(G)$ provably better than the Lovász number $\vartheta(G)$ and, in turn, $\vartheta(G)$ bounds $\alpha(G)$ from above not provably better than the clique partition number $\bar{\chi}(G)$.*

Indeed, any such upper bound on $\alpha(G)$ allows for polynomial time recognition of $\bar{\chi}(G) - \alpha(G) > 0$ gap whenever $\bar{\chi}(G)$ is known. According to Theorem 1, this would imply $P = NP$.

Acknowledgements. The first author would like to thank Günter Stertenbrink for providing him with introductory information on QCP and its 3D encoding.

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